Forecasting Durham, UK, Daily Temperature for 2020

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1. Introduction

Forecasting daily temperature is important for understanding climate variability and supporting environmental decision making. This study forecasts daily temperature for 2020 using the Durham Observatory record (1901 to 2019). Three approaches are compared: seasonal naive, ARIMA, and Fourier-based ARIMA. Performance is evaluated against 2020 observations (Durham University, 2020) using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2. The objective is to identify the most accurate forecasting approach.

2. Exploratory Analysis

The analysis used daily Durham Observatory temperature data from 1901 to 2019. Exploratory analysis identified the main characteristics of the series. The series is highly variable and dominated by a clear annual cycle (*Image 1*). Temperatures are higher mid-year and lower at the beginning and end of each year.

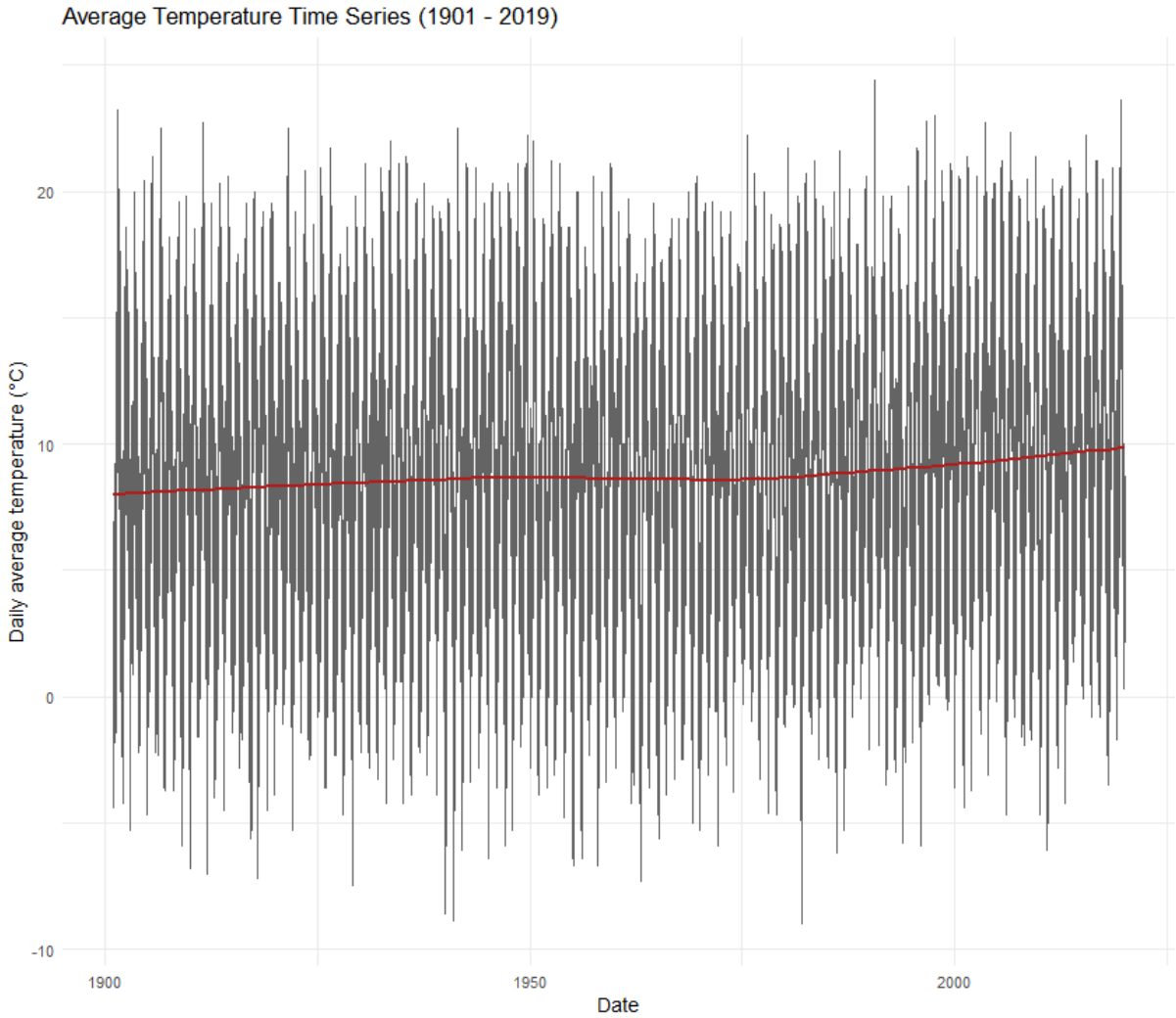

Image 1: Average Temperature Time Series from 1901 to 2019 in °C

To further examine the seasonal structure, monthly boxplots were constructed (*Image 2*). The boxplots show increasing median temperatures from winter to summer and decreasing thereafter, confirming strong seasonality.

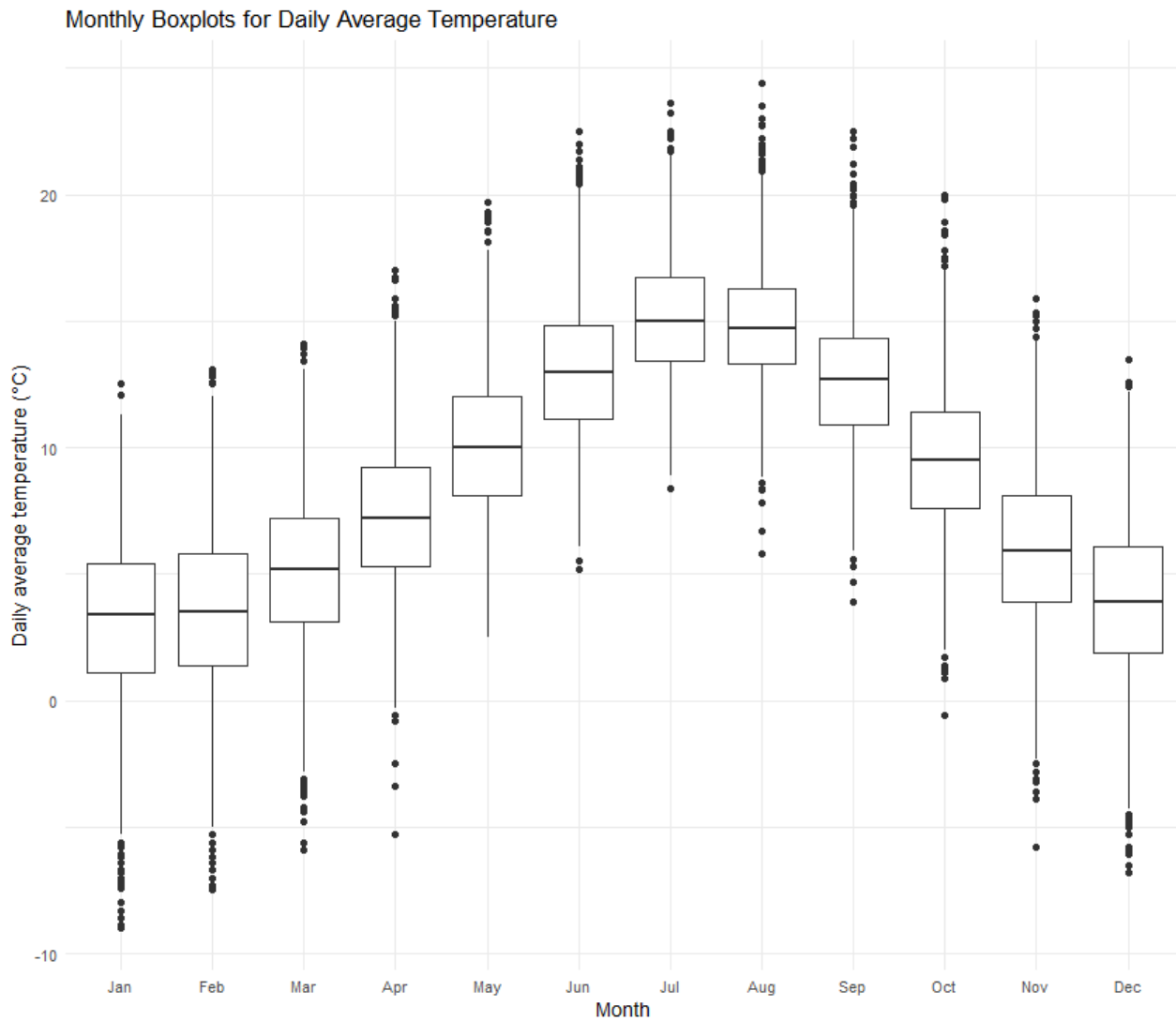

Image 2: Monthly Boxplots for the Daily Average Temperature

A 30-day moving average smooths short-term fluctuations while still reflecting seasonal variability, whereas a 365-day moving average reveals a gradual upward trend in average temperature over the full observation period, suggesting a long-term warming signal (*Image 3*). However, compared to the magnitude of seasonal variation, this trend is relatively small.

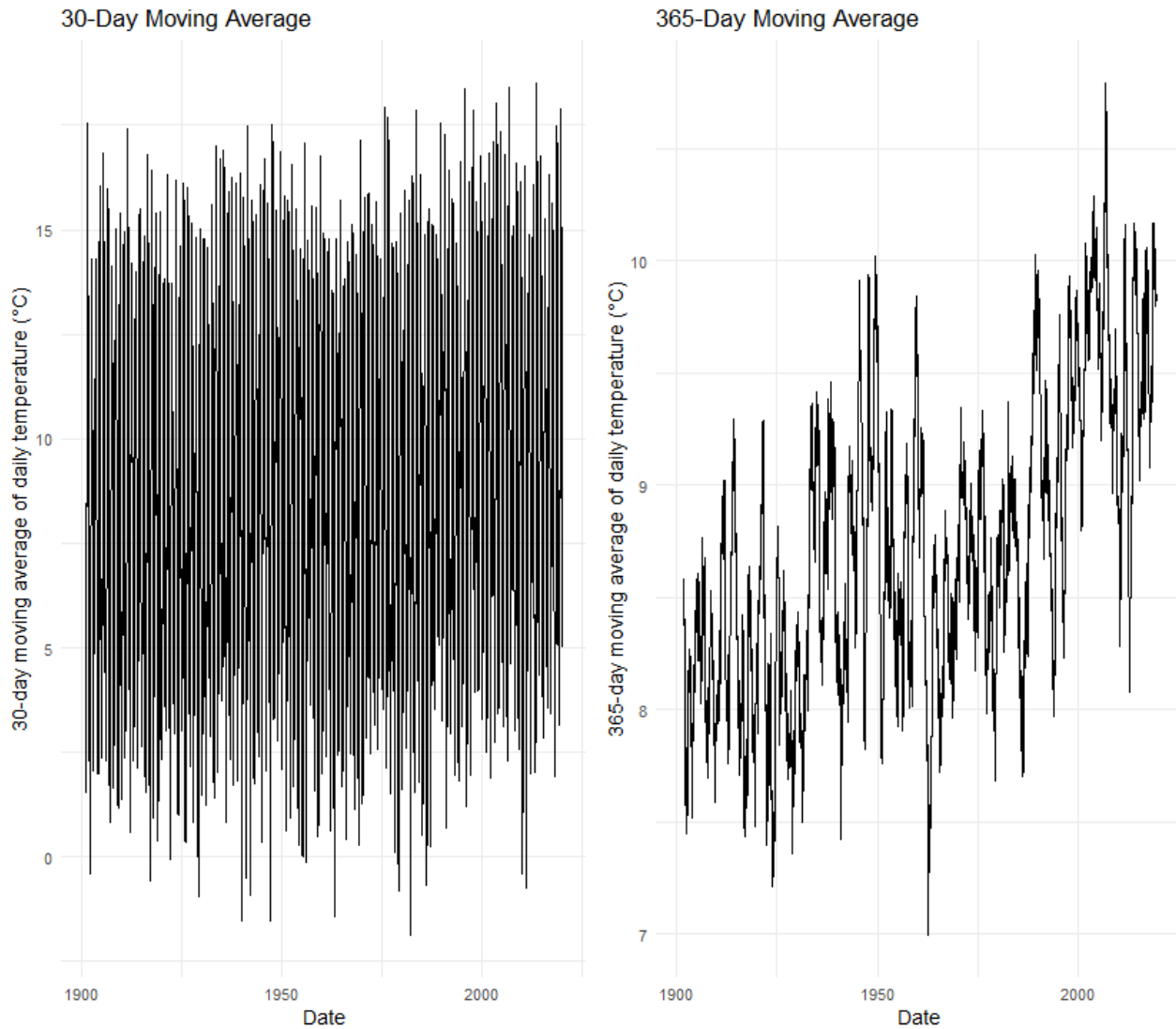

Image 3: Long-Term Behavior Analysis Using Moving Averages

The series is characterised by strong seasonality, short-term variability, and a weaker long-term trend. Seasonality is the dominant component and must be explicitly modelled, guiding the choice of forecasting methods.

3. Trend and Seasonal Diagnostics

An Augmented Dickey-Fuller (ADF) test was performed to evaluate stationarity. The ADF test rejects the unit root hypothesis ($p < 0.01$), indicating no unit-root (Hyndman & Athanasopoulos, 2014, pp. 220-221). However, strong seasonality remains, so the series is not suitable for modelling without transformation. To assess long-term trend, a linear model was fitted using time as a predictor. The linear trend reduces variance slightly (25.98 to 25.81), confirming a weak trend relative to seasonality. Finally, seasonality was estimated using median daily temperature and removed by subtraction. Variance reduced to 7.55, confirming that most variability is seasonal. Also, the Fourier power spectrum confirms strong annual periodicity (*Appendix 1*). The ACF shows short-lag dependence, and the PACF suggests a low-order

autoregressive structure (*Appendix 2 and 3*). These results justify models that separate seasonal and stochastic components.

4. Forecasting Models

These models were selected to compare different approaches for handling seasonality and assess whether modelling seasonal structure improves predictive accuracy.

4.1. Seasonal Naive

A seasonal Naive method was used as a benchmark model, where forecasts equal the value from the previous seasonal period (Makridakis et al., 2020). This assumes that the seasonal pattern is perfectly persistent over time, and uses the values from 2019 to predict the values of 2020. It serves as a baseline but cannot capture deviations or trends.

4.2. ARIMA deseasoned

ARIMA models combine autoregressive and moving average components with differencing to handle non-stationarity (Box et al., 2016, p.177). Seasonality was removed and the residual series modelled using ARIMA (5,1,0) process, indicating an autoregressive structure of order five (p = 5) and first order differencing (d=1) to remove residual non-stationarity. Forecasts were obtained by predicting the deseasoned component and adding back the seasonal cycle. This follows the framework for modeling stationary stochastic processes by Box et al. (2016, pp. 179–182).

4.3. Fourier-based ARIMA

Seasonality was modelled using Fourier terms included as regressors in an ARIMA model. The number of Fourier terms was selected using AIC minimisation, resulting in K=2. Fourier terms approximate seasonal patterns using sine and cosine functions, which requires fewer parameters than traditional approaches (De Livera et al., 2010) This provides a smooth representation of the seasonal cycle while ARIMA captures remaining autocorrelation.

5. Model Evaluation Using 2020 Observed Data

Performance was evaluated using RMSE, MAE and R^2 (*Table 1*). The ARIMA deseasoned model achieved the best performance by separating seasonal and stochastic components. This allows it to capture short-term variability more effectively while retaining the seasonal structure (*Image 4 - A and Appendix 4*). The seasonal naive model assumes perfect repetition from the previous year and performs worst (*Image 4 - C*). Fourier-based ARIMA produces smoother forecasts, but misses short-term fluctuations (*Image 4 - B*). This confirms that modelling seasonality improves predictive performance over naive approaches.

Model	RMSE	MAE	R^2
ARIMA (deseasoned)	2.35	1.91	0.73
Fourier + ARIMA	2.42	1.93	0.72
Seasonal Naive	3.65	2.92	0.50

Table 1: Model Comparison

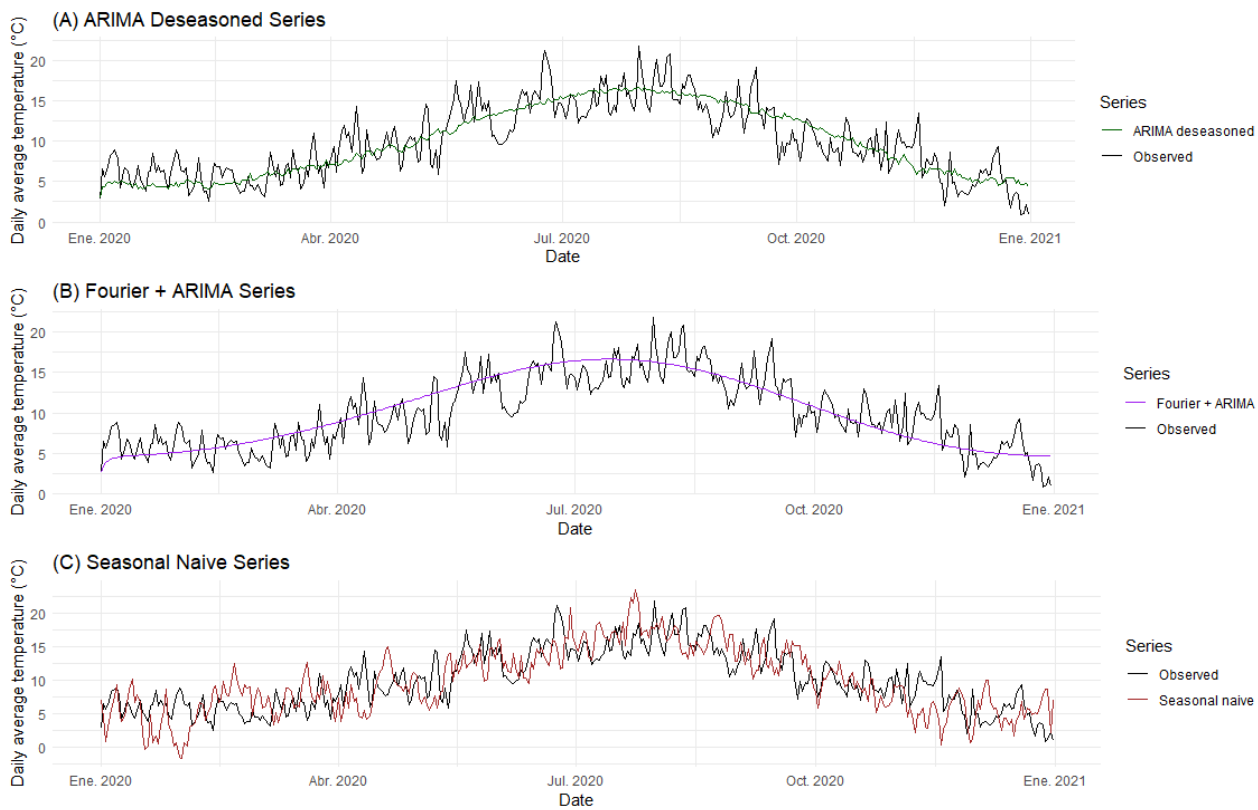

Image 4: Observed vs Predicted Values

6. Limitations

The ARIMA deseasoned approach has several limitations. Seasonality is assumed constant over time, which may not hold under climate change. Diagnostic tests show residual autocorrelation remains as shown by the Ljung-Box test, indicating incomplete capture of temporal dependence (*Appendix 5*). Additionally, all models are univariate and exclude external factors. The ARIMA model also underestimates extreme values, as shown by a regression slope lower than one (*Appendix 4*).

7. Improvements

Future work could include Seasonal ARIMA (SARIMA) or additional Fourier terms. Incorporating more predictor variables such as atmospheric pressure, humidity, precipitation, or

other atmospheric variables may improve accuracy. Further work could also focus on modelling extreme values using specialised methods.

8. Conclusion

This study forecasts daily temperature for 2020 using data from 1901 to 2019. Strong annual seasonality was identified as the dominant feature, with weaker long-term trend and short-term variability.

The ARIMA deseasoned model performed best, achieving the lowest RMSE and MAE. By separating seasonal and stochastic components, it captures short term dependence more effectively than alternative approaches.

Overall, explicitly modelling these components improves forecast accuracy relative to naive and smoothed methods. However, the model underestimates extreme values and leaves residual autocorrelation, suggesting that further improvements could be achieved through more flexible seasonal modelling or additional predictors.

9. Bibliography

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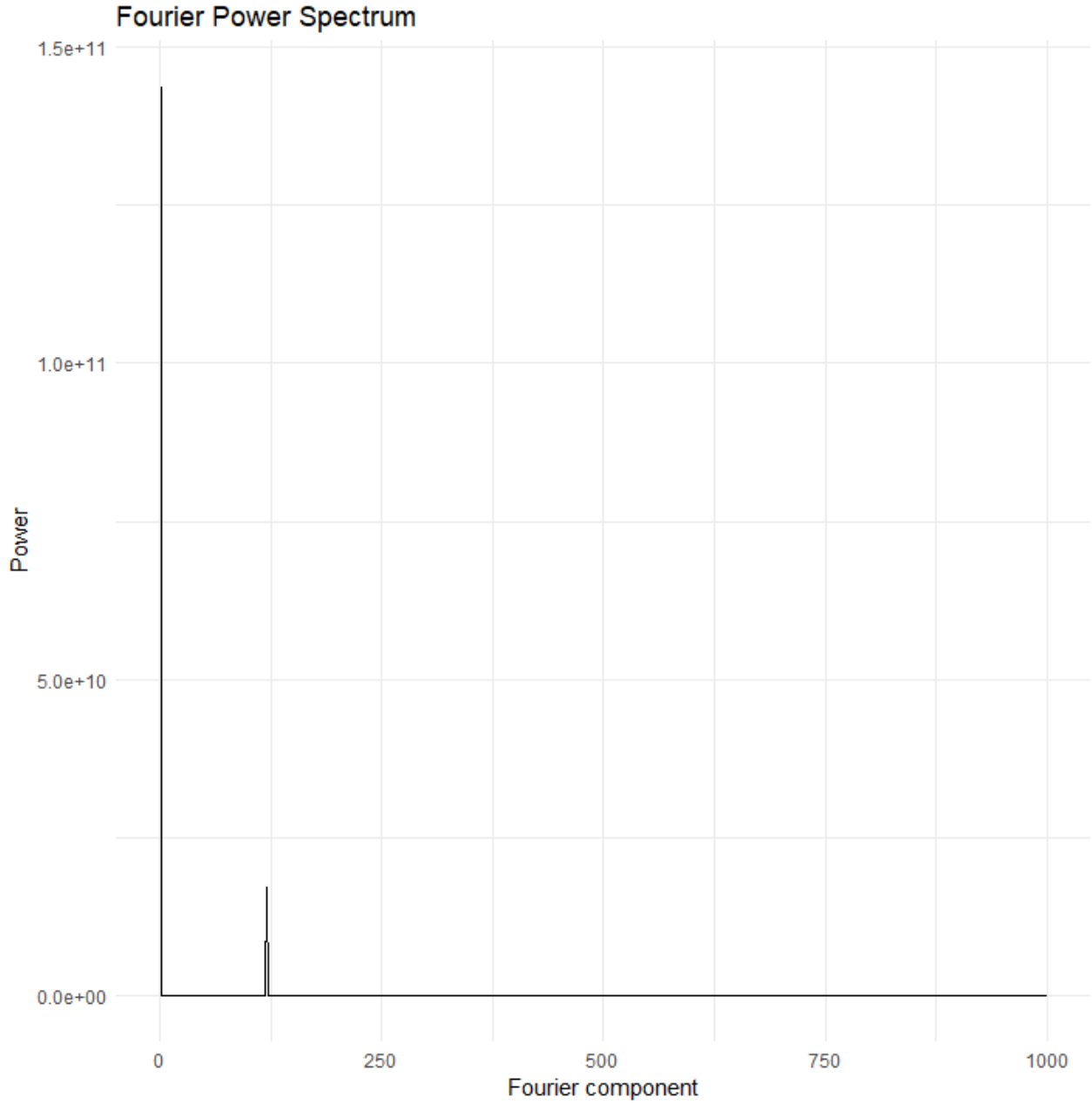
Hyndman, R. J., & Athanasopoulos, G. (2014). Forecasting: Principles and practice. OTexts. https://pdfcoffee.com/qdownload/forecasting-principles-and-practice-rob-pdf-pdf-free.html

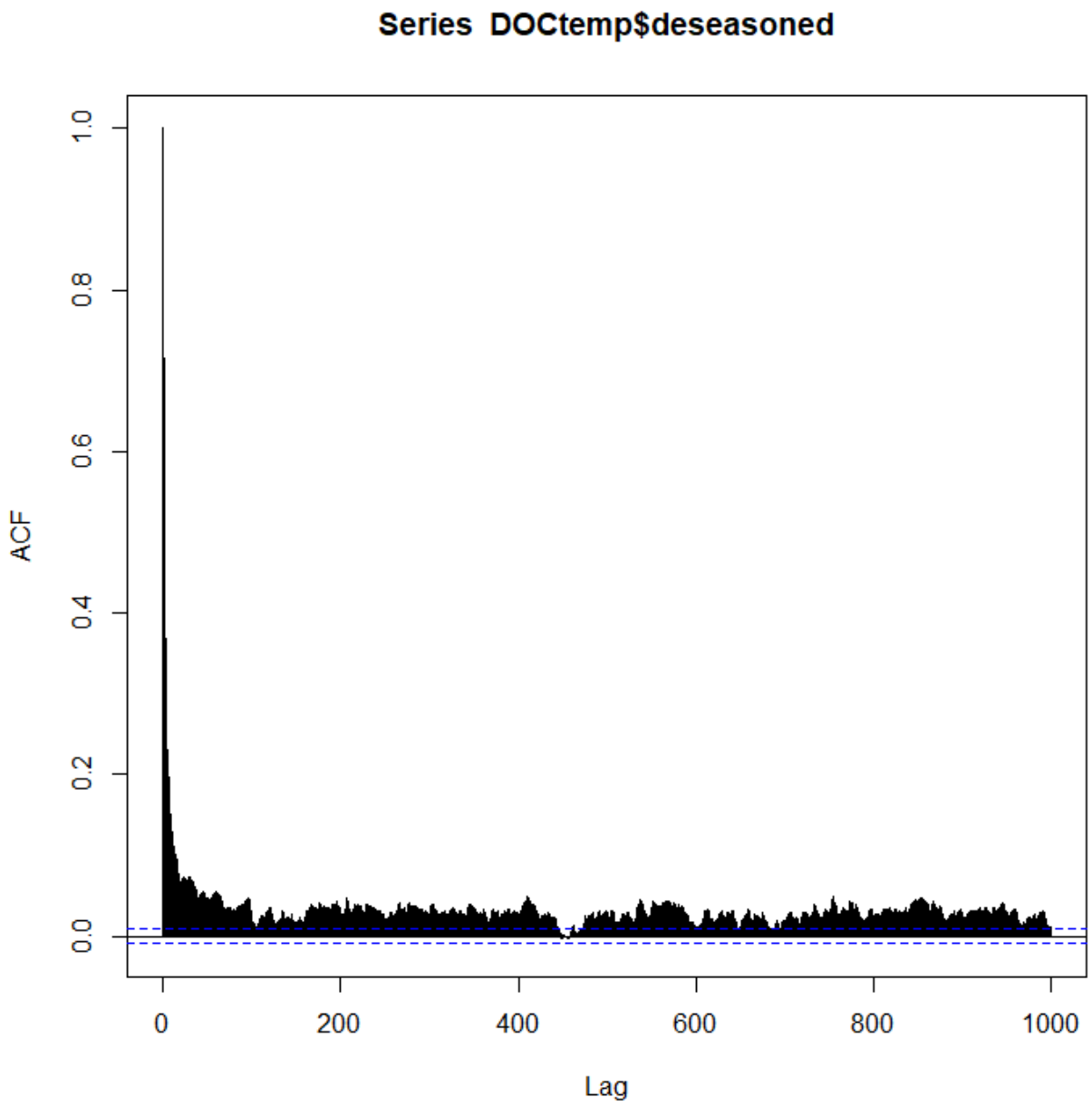
Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). The M4 Competition: Results, findings, conclusion and way forward. *International Journal of Forecasting*, 36(1), 54–74. https://www.sciencedirect.com/science/article/pii/S0169207019301128

10. Appendix

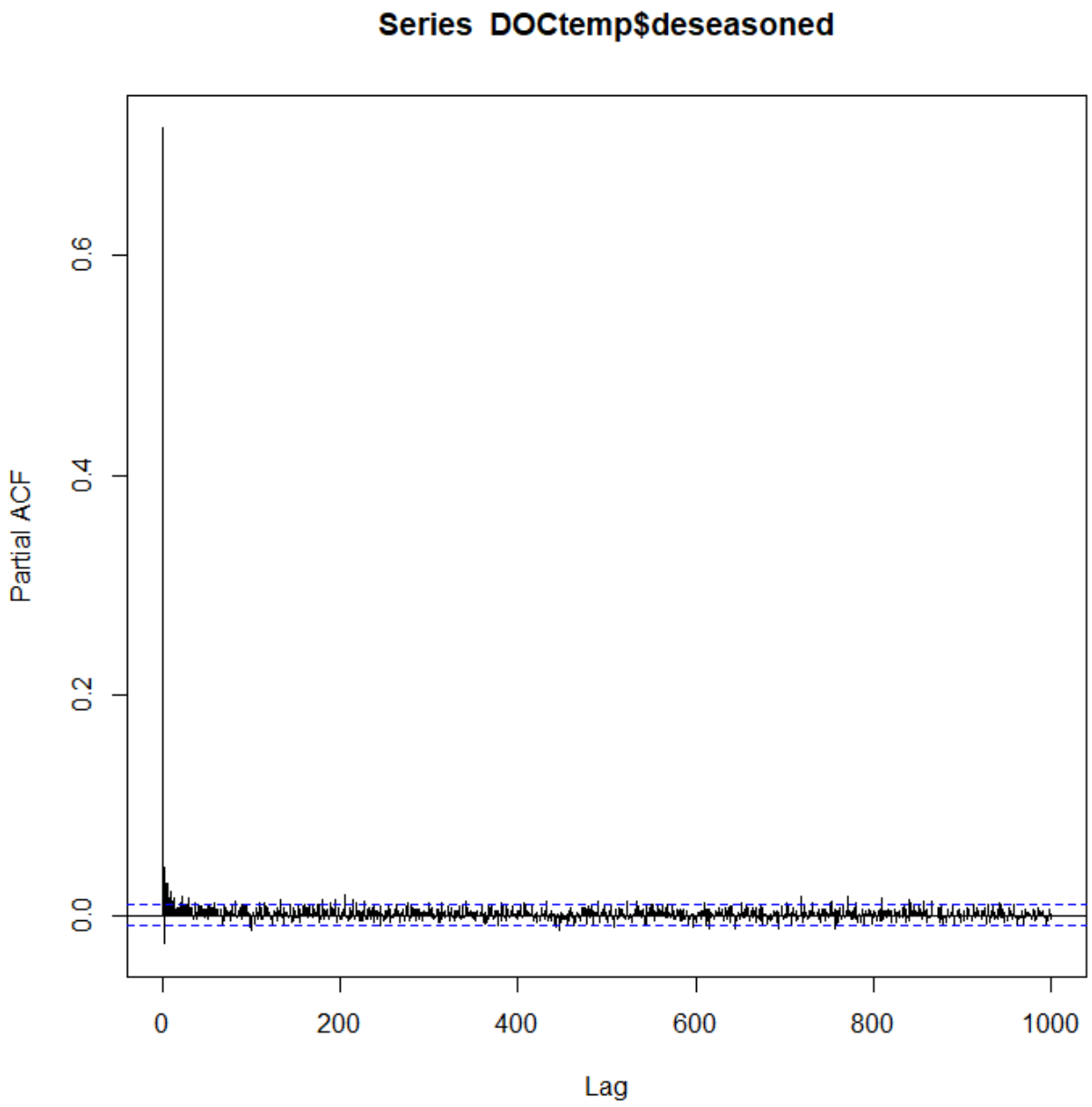
Appendix 1: Fourier Power Spectrum



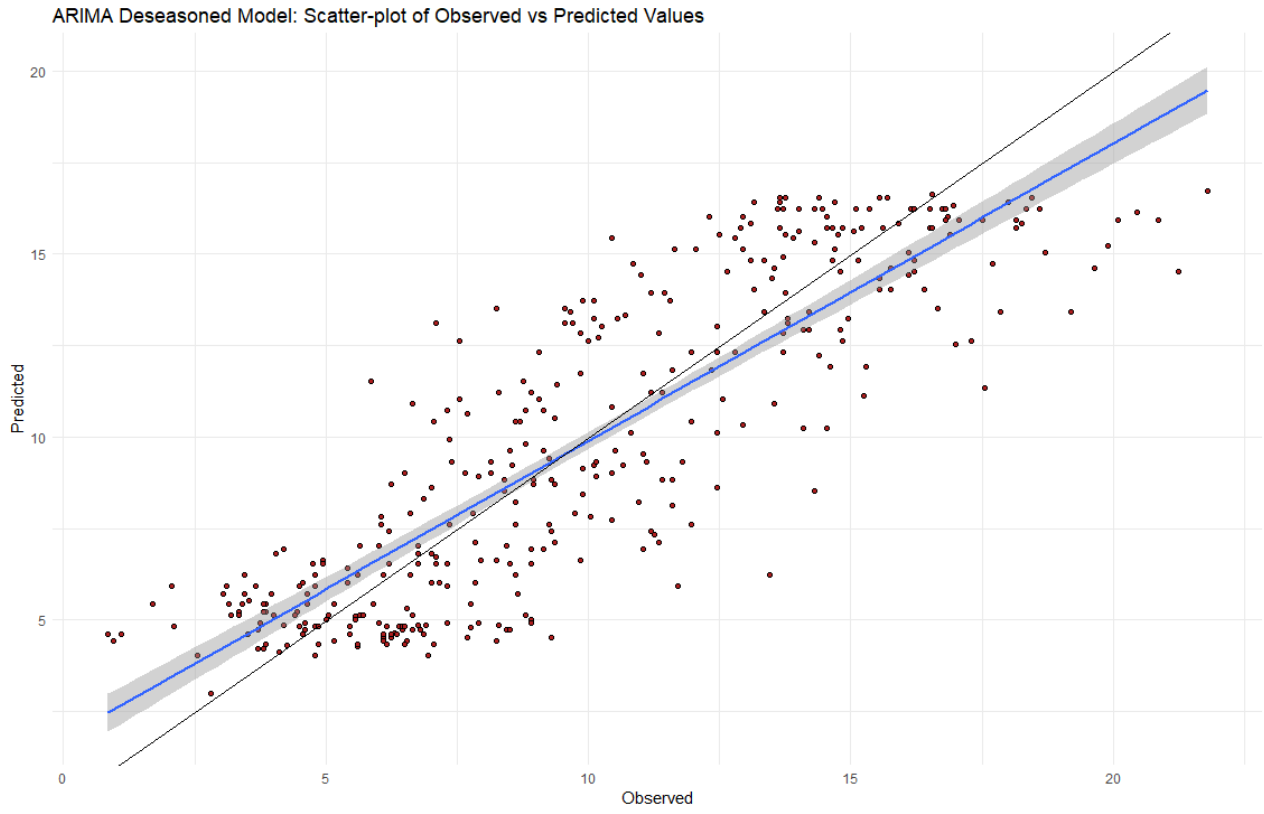
Appendix 2: Autocorrelation Function

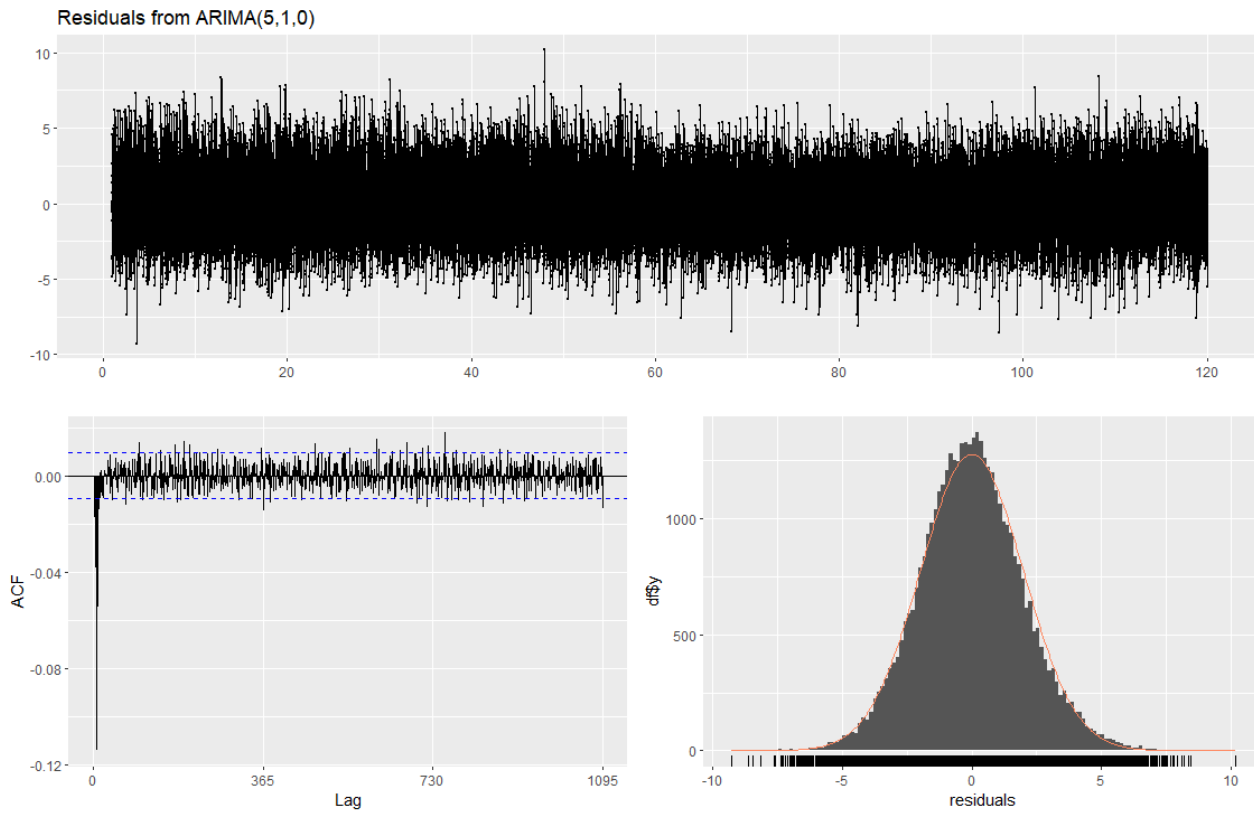


Appendix 3: Partial Autocorrelation Function



Appendix 4: ARIMA Deseasoned Scatter-plot of observed vs predicted Values



Appendix 5: Residuals ARIMA Deseasoned Model

```
> checkresiduals(model_arima_deseasoned)

        Ljung-Box test

data:  Residuals from ARIMA(5,1,0)
Q* = 2061.5, df = 725, p-value < 2.2e-16

Model df: 5.   Total lags used: 730
```